

MATH 3312, SPRING 2026: HOMEWORK 10

- (1) Let R be a PID and let $q_1, \dots, q_r \in R$ be irreducible elements generating *distinct* prime ideals. Suppose that, for $i = 1, \dots, r$, we have integers $m_i, n_{i,j} \geq 0$ for $j \leq m_i$ and set $M = \prod_{i=1}^r \prod_{j=1}^{m_i} [R/(q_i^j)]^{n_{i,j}}$.
- (a) For any irreducible element $q \in M$ and any $n \geq 1$, give a formula for $\text{length}(q^n \cdot M)$ in terms of the product decomposition.
 - (b) Conclude that the prime ideals $(q_1), \dots, (q_r)$ and the integers m_i and $n_{i,j}$ depend on M only up to isomorphism.
 - (c) Conclude that the invariant factors of a finitely generated R -module are indeed uniquely determined up to invertible elements.
- Hint: For the uniqueness, see the argument in Lecture 37 from 3311.*

- (2) Show that $\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subset \mathbb{R}$ is Galois over $\mathbb{Q}$ and compute its Galois group.

Let $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$ be a *primitive* n -th root of 1. Define

$$\Phi_n(x) = \prod_{j \in (\mathbb{Z}/n\mathbb{Z})^\times} (x - \zeta_n^j) \in \mathbb{C}[x].$$

This is the **n -th cyclotomic polynomial**.

- (3) When $n = p$ is prime, show that $\Phi_p(x) = x^{p-1} + \dots + x + 1$, and use Eisenstein's criterion to conclude that $m_{\zeta_p}(x) = \Phi_p(x) \in \mathbb{Q}[x]$.
- (4) Show that $\mathbb{Q}(\zeta_p) \subset \mathbb{C}$ is Galois over $\mathbb{Q}$ with Galois group isomorphic to $(\mathbb{Z}/p\mathbb{Z})^\times$.
- (5) Suppose that we have a Galois extension L/k , and that K/k is a k -sub-extension of L . Given $\sigma \in \text{Gal}(L/k)$, show that

$$\text{Gal}(L/\sigma(K)) = \sigma \text{Gal}(L/K) \sigma^{-1}.$$

Conclude that we have $\sigma(K) = K$ for all $\sigma \in \text{Gal}(L/k)$ if and only if $\text{Gal}(L/K) \trianglelefteq \text{Gal}(L/k)$ is a normal subgroup.

Hint: We will see in class that L is Galois over K , so that $\text{Gal}(L/K)$ makes sense. Think of $\text{Gal}(L/K)$ as the simultaneous stabilizer in $\text{Gal}(L/k)$ of all the elements in K .

- (6) Suppose that p is odd, and let $H \leq (\mathbb{Z}/p\mathbb{Z})^\times$ be the unique subgroup of order 2. Compute the fixed field

$$\mathbb{Q}(\zeta_p)^H = \{\alpha \in \mathbb{Q}(\zeta_p) : \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}.$$

- (7) Show that $\mathbb{Q}(\sqrt[4]{2}, i)$ is Galois over $\mathbb{Q}$, and compute its Galois group. If you're feeling ambitious, write down all of its subextensions as well, or at least figure out how many such subextensions there are.